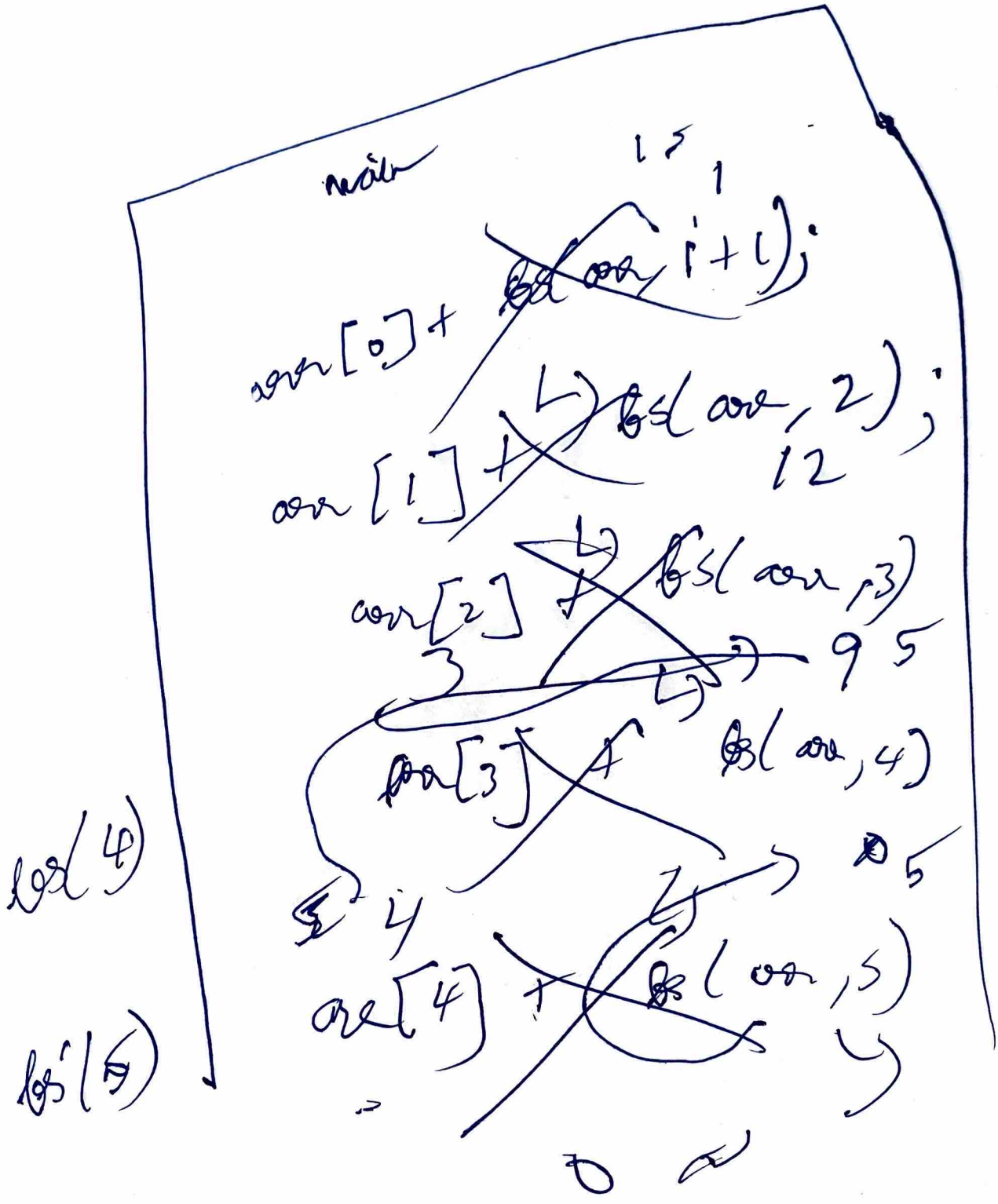
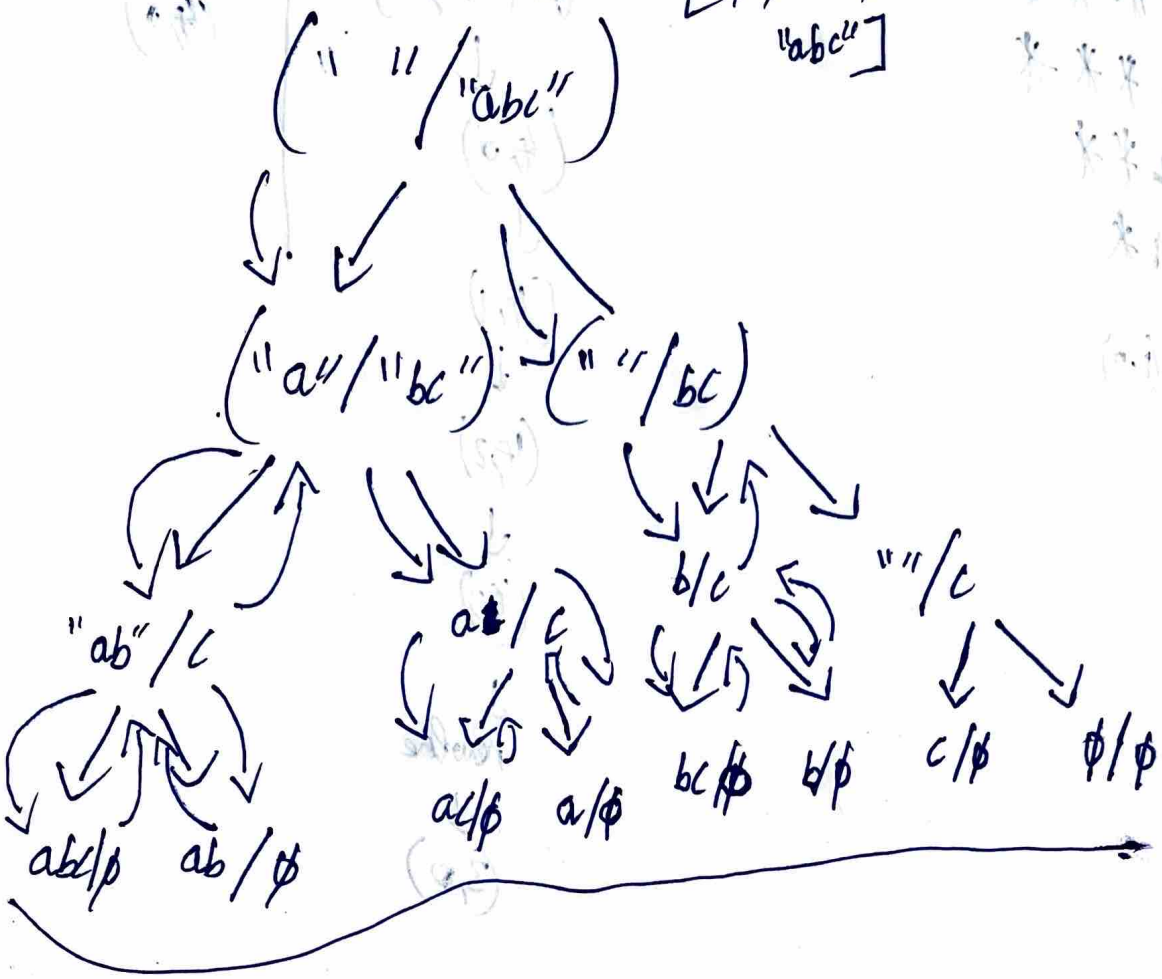

main
arr[0] +
arr, i+1);
arr[1] +
fs(arr, 2);
arr[2]
fs(arr, 3)
arr[3]
fs(arr, 4)
arr[4] +
fs(arr, 5)
fs(4)
fs(5)

abc

["a", "ab", "b.c", "ac", "b", "c", "abc"]



[1, 2, 3]

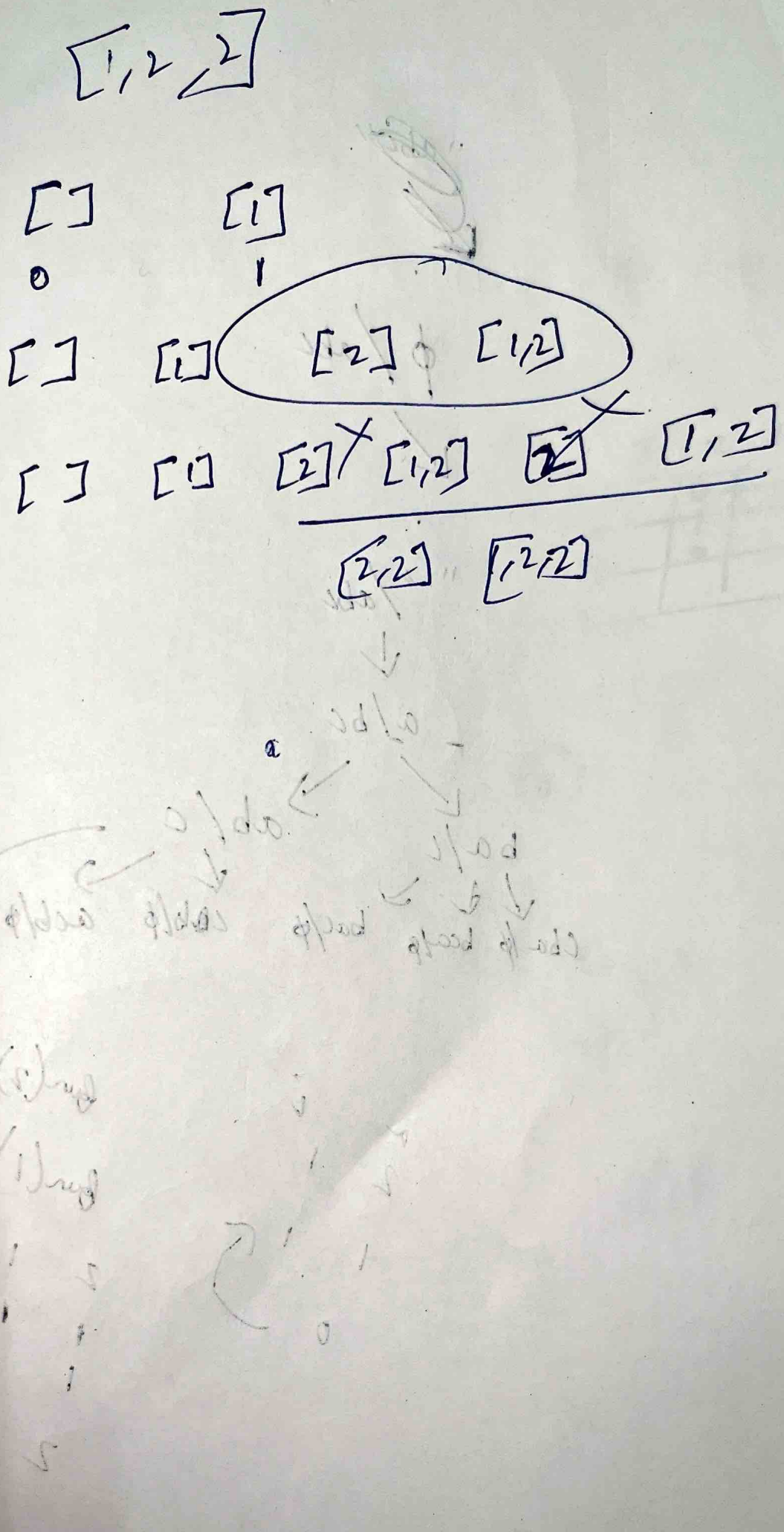
[ ] [1]

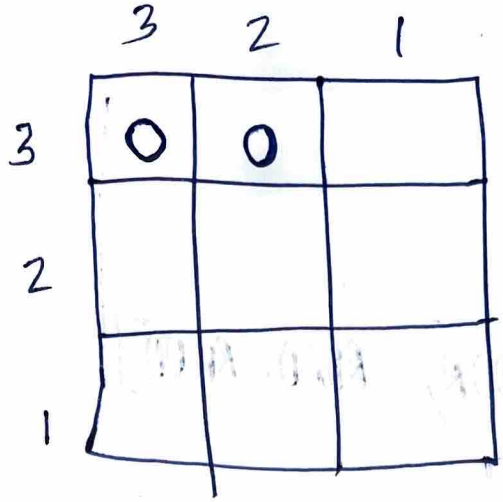
[ ] [1] | [2] [1,2]

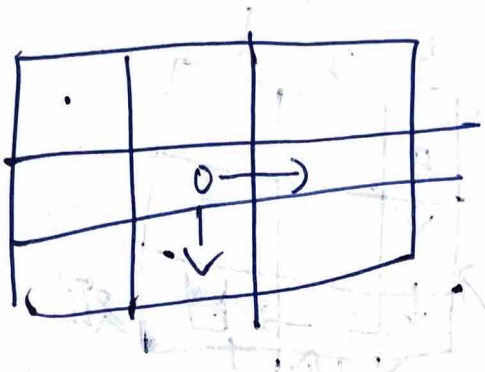
[ ] [1] [2] [1,2] | [3] [1,3] [2,3]

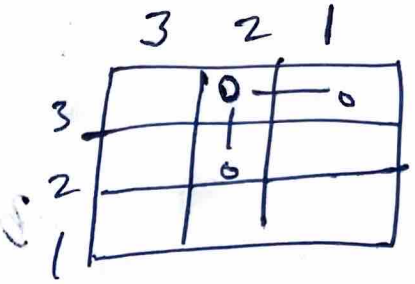
[1,2,3]

Ans

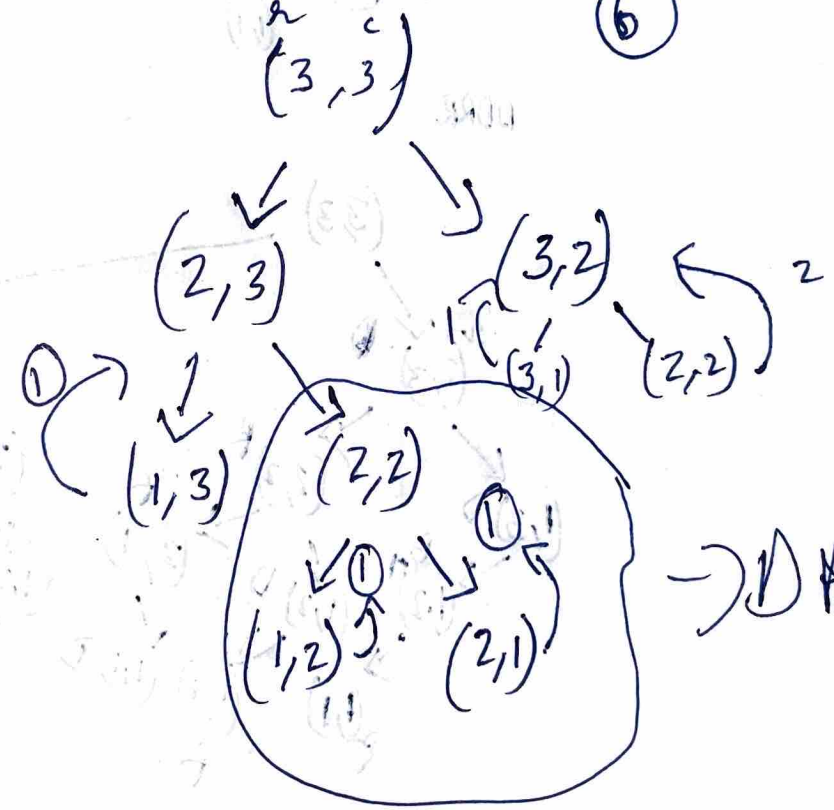

[1,2,2]
[]
[1]
0
1
[]
[1]
[2]
[1,2]
[]
[1]
[2]
[1,2]
[1,2]
[2,2]
[1,2,2]



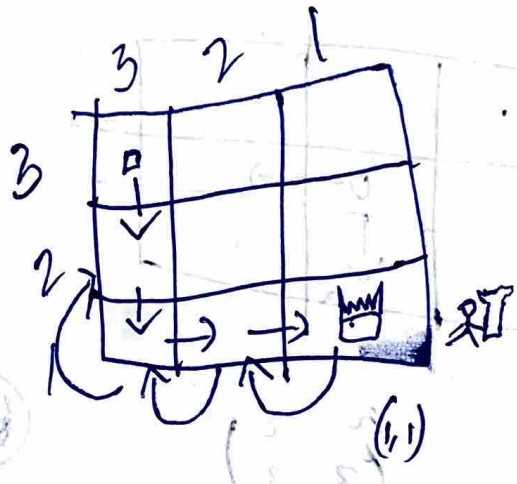




(6)

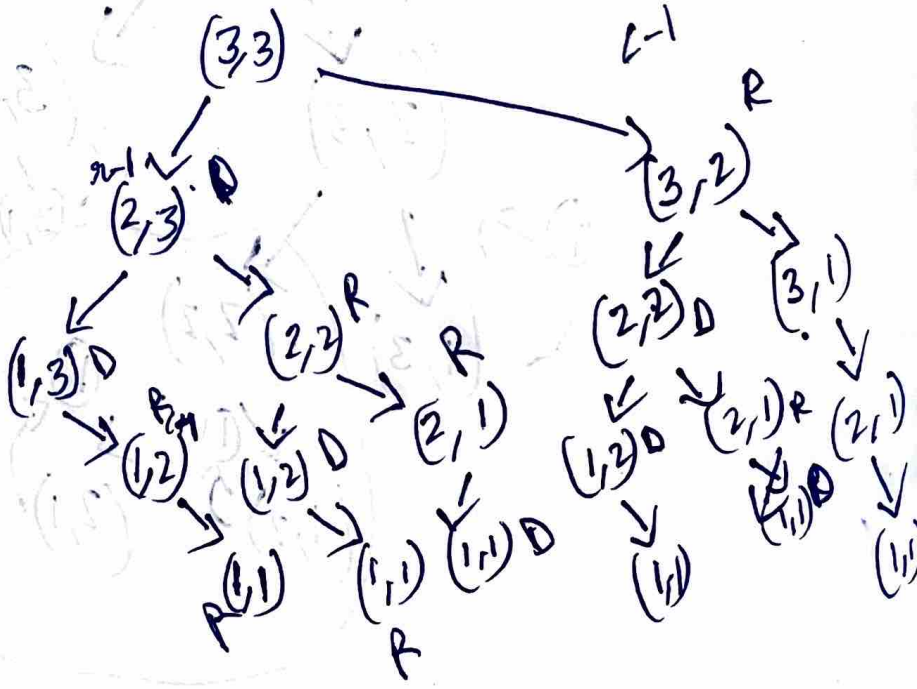


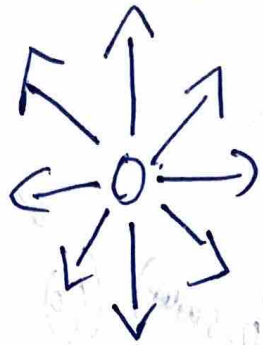
[DDRR, DROR, DRRD, RDDR, RDRD, RROO]





DDRR

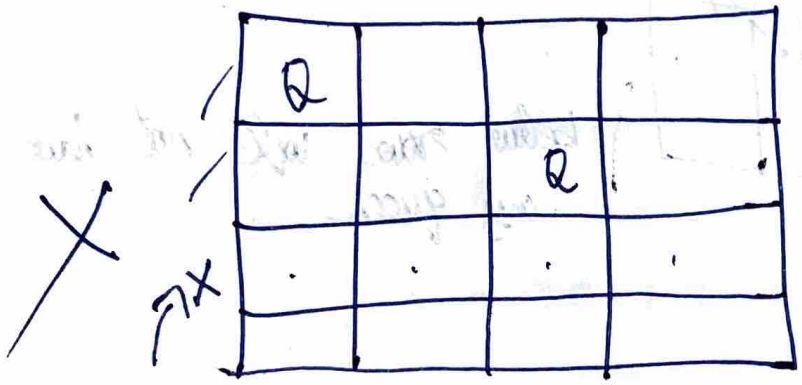




n=4

n x n matrix

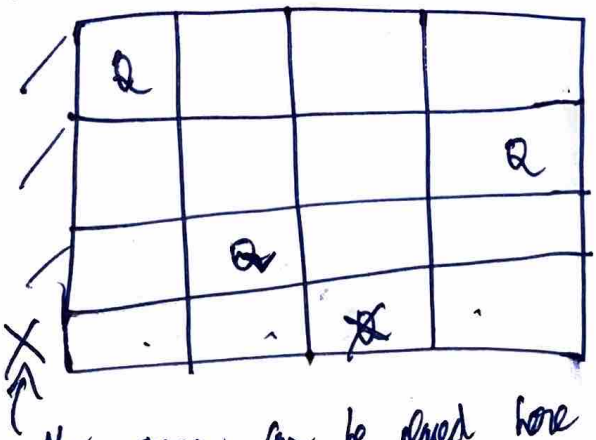
Place 4 Queens on it



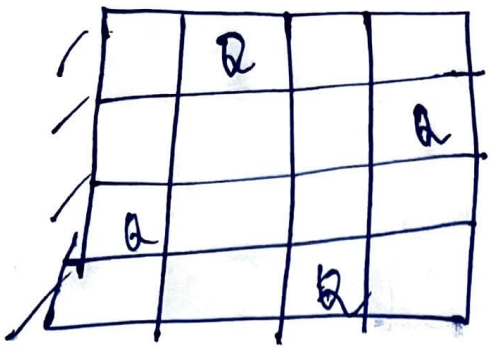
How I know it is recursion

backtracking?

No way you can add queen here



No queen can be placed here



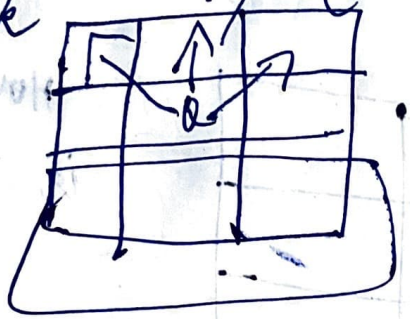
(board, 0, C)

one of the answer

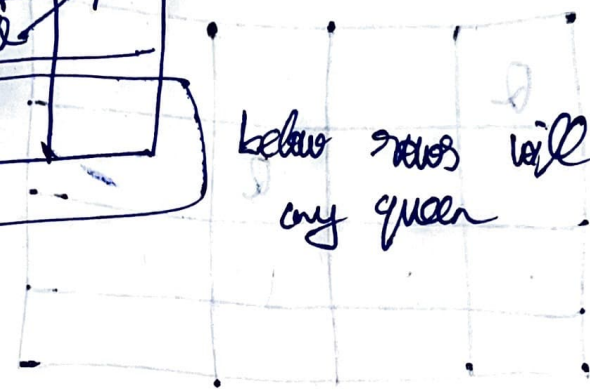
(0,0) (0,1) (0,2)

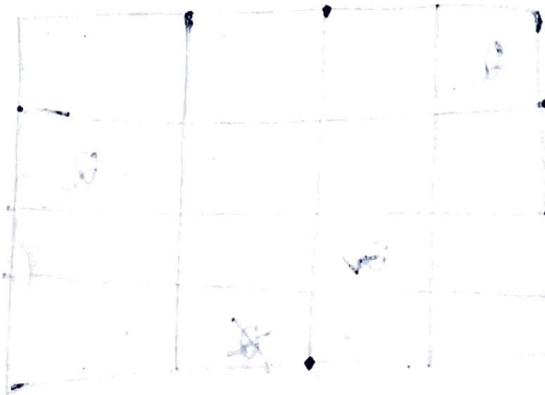
(1,0) (1,1) (1,2)

to check



below rows will not have any queen





0	1	1	2	3	5	8	13	21
[illegible]	2	3	4	5	6	7	8	9
0	1	2	3	4	5	6	7	8

$f(4) = f(3) + f(2)$

4

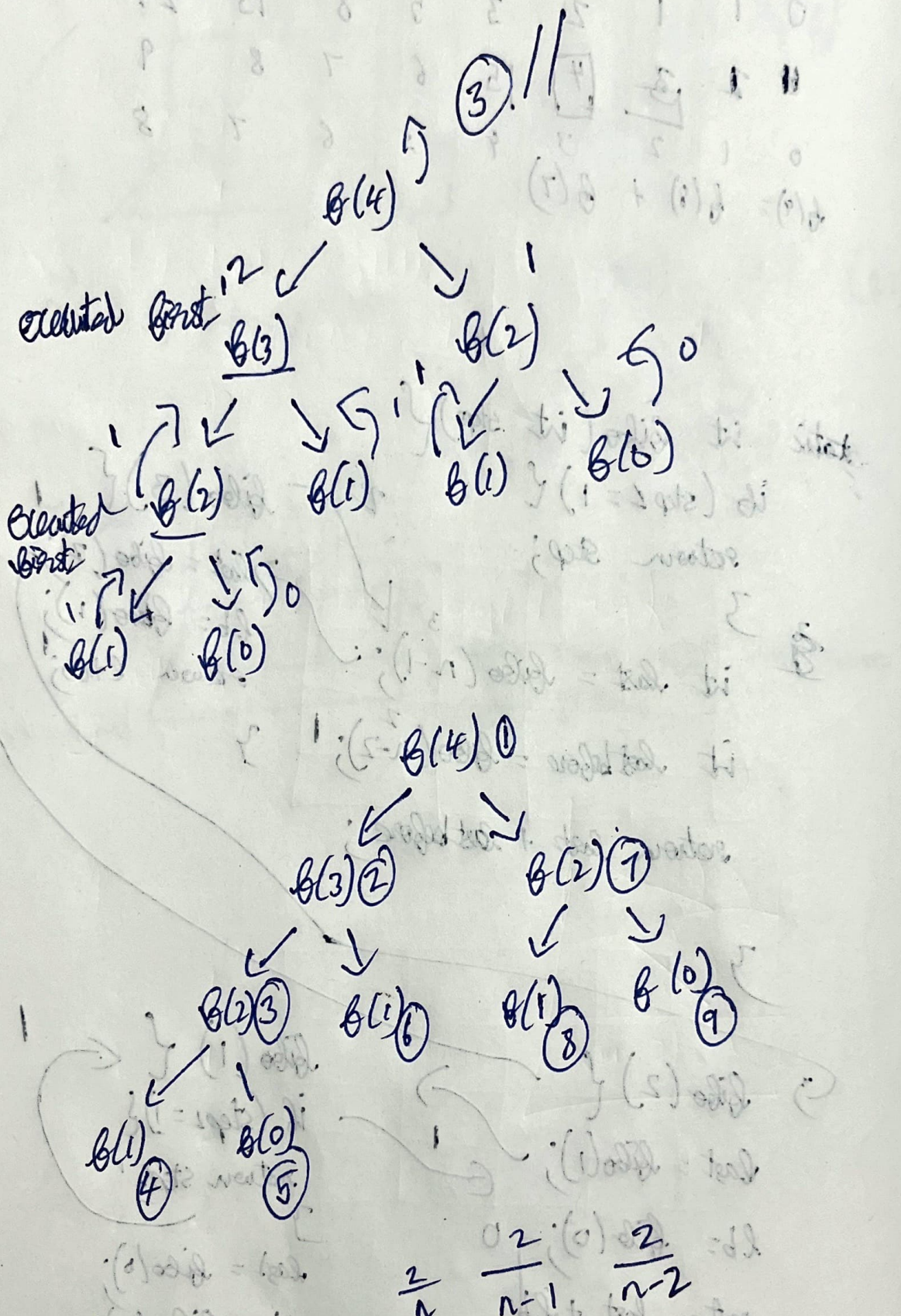
```
static int fibo(int step){
    if (step <= 1){
        return step;
    }
    int last = fibo(n-1);
    int lastbefore = fibo(n-2);
    return last + lastbefore;
}
```

```
fibo(3){
    last = fibo(2);
    lb = fibo(1);
    return l+lb;
}
```

```
fibo(2){
    last = fibo(1);
    lb = fibo(0);
    return last + lb;
}
```

```
fibo(1){
    if (step <= 1){
        return step;
    }
    last = fibo(0);
    lb = fibo(-1);
    return
```

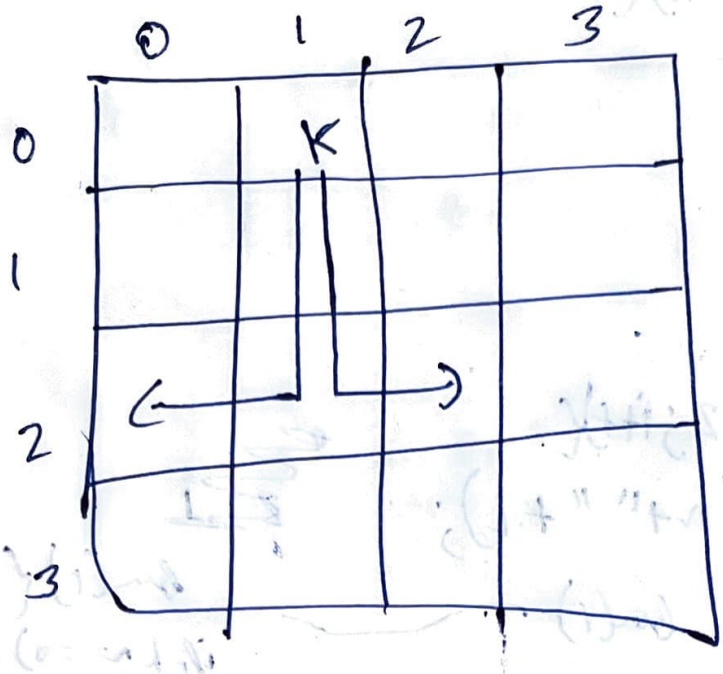
```
fibo(0){
    if (step <= 1){
```



$\frac{2}{n}$ $\frac{2}{n-1}$ $\frac{2}{n-2}$

$T(n) \to (2^n)$

exponential in nature



(0, 1)

```
[ row-2, col-1
  row-2, col+1
  row-1, col-2
  row-1, col+2 ]
```